

DSM MINI PROJECT

10. DC Motor

A. Electrical Equation (KVL)

$$V = L \frac{di}{dt} + Ri + k_e \omega$$

L, R : Inductance and Resistance

B. Mechanical Equation (Newtonian)

$$J \frac{d\omega}{dt} = T_m - D\omega - T_L$$

$T_m = k_t i$: Motor torque

D : Damping / friction coefficient

T_L : External load Torque

C. Solving Differential Equation

$$\frac{d\omega}{dt} + \frac{D}{J} \omega = \frac{k_t i}{J}$$

~~IF~~ Using integrating factor

$$\omega(t) = \frac{k_t i}{D} (1 - e^{-\frac{D}{J} t})$$

2. Stepper Motor

A. Torque Equation

T_s is proportional to displacement θ_d

$$T_s = -T_{max} \sin(N(\theta - \theta_d))$$

T_{max} : Holding Torque

N : No. of poles

B. A Equation of Motion

$$J \frac{d^2\theta}{dt^2} + D \frac{d\theta}{dt} = T_s - T_L$$

$$J\ddot{\theta} + D\dot{\theta} + (T_{max}N)\theta = (T_{max}N)\theta_d$$

3. Servo Motor Derivation

A. Control Law (Torque Generation)

$$e = \theta_{ref} - \theta$$

$$T_c = k_p e + k_i \int e \, dt + k_d \frac{de}{dt}$$

B. Dynamic Equation

Substituting Controller output into the motion equation

$$J \frac{d\omega}{dt} = T_c - D\omega$$

State - Space form

$$x_1 = i, x_2 = \omega$$

$$\dot{x}_1 = -\frac{R}{L}x_1 - \frac{k_e}{L}x_2 + \frac{1}{L}V$$

$$\dot{x}_2 = \frac{k_t}{J}x_1 - \frac{D}{J}x_2$$

Matrix form ($\dot{x} = Ax + Bu$)

$$\begin{bmatrix} \dot{i} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} -R/L & -k_e/L \\ k_t/J & -D/J \end{bmatrix} \begin{bmatrix} i \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} V$$

Unit check

Torque (T) : $\text{N} \cdot \text{m}$

Inertia (J) : $\text{kg} \cdot \text{m}^2$

Current (i) : Amperes

Angular Velocity (ω) : rad/s

Resistance (D) : $\text{N} \cdot \text{m} \cdot \text{s/rad}$